



COVID-19 Vaccination Trends and Global Health Inequality

How do vaccination trends differ across countries, and what do they reveal about global health inequality?

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Key highlights

- ☀️ Only 61 of 216 countries (28.2%) reached 50% full vaccination coverage by September 2021. The remaining 155 countries 71.8% of the world remained below this threshold throughout the entire study period.
- ☀️ The gap between income groups widened continuously throughout 2021. High-income countries reached approximately 50 fully vaccinated per 100 inhabitants by September 2021, while low-income countries remained near zero a difference that accelerated from April 2021 onwards.
- ☀️ The difference between the highest and lowest observed vaccination coverage exceeded a factor of 11,000. Gibraltar reported 116.4 fully vaccinated per 100 inhabitants; the Democratic Republic of Congo reported 0.01.
- ☀️ Among the 61 countries that reached 50% coverage, the median rollout time was 160 days. The fastest countries achieved this in under 30 days; the slowest required 235 days. Speed, not just coverage, was unequally distributed.
- ☀️ Missing data is not random. Countries with near-complete missingness ($\geq 95\%$) are overwhelmingly lower-income and lower-capacity nations. Tokelau with 100% missing records nonetheless achieved 96% vaccination coverage, demonstrating that absence of data does not mean absence of a campaign.
- ☀️ PCA and k-means clustering confirm that country vaccination profiles fall into three structurally distinct groups, not a continuous spectrum. Global health inequality in vaccination was categorical, not gradual.
- ☀️ ARIMA forecasting reveals a further dimension of inequality: Germany's 30-day forecast carries narrow confidence intervals, while Bangladesh's forecast is highly uncertain. Unreliable data makes evidence-based planning harder precisely where it is needed most.

Overview

This report analyses global COVID-19 vaccination trends using the Our World in Data COVID-19 vaccination dataset (Mathieu et al., 2021), covering 222 countries and territories between December 2, 2020, and August 28, 2021. Using descriptive statistics, principal component analysis, k-means clustering, and ARIMA time series forecasting, the analysis quantifies differences in vaccination coverage, rollout speed, and data-reporting completeness across countries and World Bank income groups.

Only 28.2% of countries reached 50% full vaccination coverage during the observation period, and coverage differed by more than 50 percentage points between the highest- and lowest-income groups. A forecasting comparison across Spain, India, and Bangladesh further shows that the reliability of short-term vaccination forecasts is itself unequally distributed once appropriately scaled for each country's own coverage level and extending the inequality finding beyond outcomes and into the data infrastructure used to monitor them.

Keywords: COVID-19, vaccination, global health inequality, principal component analysis, k-means clustering, ARIMA forecasting

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1. Introduction

The COVID-19 pandemic presented an unprecedented challenge to global public health systems. The rapid development and subsequent worldwide deployment of COVID-19 vaccines in 2020 and 2021 represented a landmark achievement in medical science, but the distribution and administration of those vaccines across countries was deeply uneven, raising fundamental questions about global health equity (Wouters et al., 2021). While early commentary on vaccine inequity relied largely on aggregate, country-level snapshots, the daily, publicly reported data compiled by Our World in Data (Mathieu et al., 2021) make it possible to examine not only how much of the world was vaccinated, but how quickly, how consistently, and how reliably that progress could be tracked and forecast in the first place.

This report presents a data-driven analysis of COVID-19 vaccination trends across 222 countries and territories during the first nine months of global vaccine deployment, from December 2020 to August 2021. The analysis is organized around a central research question: how do vaccination trends differ across countries, and what do they reveal about global health inequality? The report is structured as follows. Section 2 reviews the state of the art and defines the specific problem addressed. Section 3 states the analytical objectives. Section 4 describes the dataset and methods. Section 5 presents and interprets the results. Section 6 discusses the findings critically, including their limitations. Section 7 concludes with a summary and an outlook for further work.

2. Problem Definition

2.1 State of the Art

Vaccine inequity during the COVID-19 pandemic has been documented extensively in the scientific literature. Wouters et al. (2021) identified multiple structural barriers to equitable vaccine access, including financial constraints, manufacturing capacity, and geopolitical factors that advantaged wealthier countries in the first year of the rollout. The COVAX initiative, established to pool procurement and ensure more equitable global distribution, faced substantial and well-documented obstacles in reaching low-income countries during 2021 (Eccleston-Turner & Upton, 2021). More broadly, Acharya and Porwal (2020) showed that pre-existing structural healthcare inequalities were strong predictors of differential outcomes during the pandemic, a pattern consistent with the vaccination-specific findings in this report.

Data availability is itself a dimension of health inequality. Countries with stronger health information systems have historically been able to report disease and health-programme data continuously, while countries with weaker administrative or digital infrastructure often report sporadically or not at all (Mbousou et al., 2019). This has direct implications for monitoring vaccination campaigns and for any statistical model including a forecasting model that depends on that reported data being complete and regularly spaced in time.

2.2 Problem Statement

Despite extensive documentation of vaccine inequity at the macro level, granular, reproducible, data-driven analyses that compare vaccination trajectories across countries using a single public dataset and that treat the reliability of the data and of any resulting forecast as part of the inequality story, not just the outcomes themselves remain comparatively rare. This report addresses four specific gaps:

- ☀ How large is the quantitative gap in vaccination coverage between high-income and low-income countries, and does it widen or narrow over the observation period?
- ☀ Do differences manifest only in final coverage, or also in the speed of rollout?
- ☀ Does data missingness correlate with a country's apparent vaccination performance, and what does that imply for monitoring?
- ☀ Can time series forecasting models reliably project vaccination progress, and is forecast reliability itself unequal across countries and if so, is that inequality still present once differences in absolute coverage level are accounted for?

3. Objectives

The primary objective of this study is to characterize global COVID-19 vaccination inequality through a reproducible, quantitative analysis of a single publicly available dataset. The specific analytical objectives are:

- ☀ To describe the distribution of vaccination coverage across countries using descriptive statistics and a diverse set of visualization techniques.
- ☀ To compare vaccination rollout speed by measuring the time required for countries to reach 50% full vaccination coverage.
- ☀ To analyse income-group and continental differences in vaccination trajectories, and to quantify how reporting completeness itself varies across countries.
- ☀ To apply unsupervised learning methods (principal component analysis and k-means clustering) to identify country groupings by vaccination profile, with the number of clusters justified empirically rather than assumed.
- ☀ To evaluate the predictive reliability of ARIMA time series models for vaccination forecasting across countries at different coverage levels, and to test rather than assume whether that reliability tracks income level.

4. Methods

4.1 Dataset

The analysis uses the Our World in Data COVID-19 vaccination dataset, a plain comma-separated CSV file with daily observations for 234 locations, of which 222 are individual countries or territories and 12 are continent- or income-group aggregates (World, Europe, European Union, Asia, Africa, North America, South America, Oceania, and the four World Bank income groups). The country-level data span December 2, 2020, to August 28, 2021, and provide 14 variables per observation, including total vaccinations, people vaccinated, people fully vaccinated, and booster doses, each available as both an absolute count and a rate per 100 inhabitants.

4.2 Data Cleaning

The raw file was loaded directly as a standard comma-delimited CSV, with missing values already encoded as empty cells; no non-standard delimiter or character-encoding correction was required. Percentage-type columns were coerced to numeric type, and date strings were parsed as calendar dates. Regional and income-group aggregate rows were separated from individual country rows: aggregates were retained for the income-group comparison (Section 5.5) but excluded from every country-level analysis to avoid double-counting, leaving 222 individual countries and territories, of which 216 have at least one non-missing full-vaccination record.

4.3 Descriptive Statistics and Visualisation

Descriptive statistics were computed for the primary outcome variable, people fully vaccinated per 100 inhabitants, including the distribution's extremes and its variation across countries. A cross-sectional snapshot was constructed by retaining each country's own latest available observation; because these "latest" dates range from April to August 2021 across countries, comparisons based on this snapshot are noted as a limitation throughout (Section 6.2). All visualizations were produced using `ggplot2` (Wickham, 2016) with a consistent colour palette, together with several extension packages `ggbreak` (Xu et al., 2021) for axis-break charts, `ggdist` and `ggbeeswarm` for the raincloud plot, and `ggcorrplot` for the correlation heat map to diversify the graphical techniques used beyond what is covered in the course.

4.4 Rollout Speed Analysis

To quantify differences in vaccination speed independently of final coverage, each country was classified according to whether it ever reached 50% full vaccination coverage during the observation period. For countries that did, the number of days between their first recorded vaccination observation and the date that threshold was reached was calculated as a proxy for rollout speed.

4.5 Principal Component Analysis (PCA)

Principal component analysis (PCA) was applied to four vaccination indicators: total vaccinations, people vaccinated, people fully vaccinated (all per hundred), and daily vaccinations per million, after standardization to zero mean and unit variance. PCA reduces these correlated indicators to a small number of orthogonal components, each capturing a share of the total variance; the first two components were retained for visualization. Countries missing a value on any of the four indicators were excluded from this step and are listed explicitly in Section 5.7.

4.6 K-means Clustering

K-means clustering was applied to the same standardized indicators to identify distinct country groupings by vaccination profile. Rather than assuming three clusters from theory alone, the number of clusters was first assessed using the within-cluster sum-of-squares ("elbow") method and the average silhouette width (Rousseeuw, 1987), both computed with the **factoextra** package. $k = 3$ was retained on the basis of these diagnostics together with the theoretically expected high/moderate/low coverage segmentation. A fixed random seed was set prior to model fitting, in accordance with the minimum standard of reproducibility described by Peng (2011). Because k-means assigns cluster numbers arbitrarily, the three resulting clusters were labelled according to each cluster's own computed mean coverage (highest, moderate, lowest) rather than by their arbitrary numeric index.

4.7 ARIMA Time Series Forecasting

Autoregressive Integrated Moving Average (ARIMA) models were fitted to the daily vaccination coverage series of selected countries using the ***auto.arima()*** function from the **forecast** package (Hyndman & Athanasopoulos, 2021), which selects its own differencing order via internal unit-root tests rather than a fixed, assumed order. A weekly seasonal frequency was specified to match the data's reporting cycle. Because not every country reports on every calendar day—a real, verified gap of 10 multi-day jumps in India's series and 32 in Bangladesh's, versus none in Germany's—each country's series was first re-expressed on a complete daily calendar with missing

days linearly interpolated (Zeileis & Grothendieck, 2005) before model fitting, so that the weekly seasonal index in the model corresponds to actual calendar days rather than to raw row position.

Germany's series was modelled first as a single-country case study, with residual diagnostics (Ljung-Box test), an ETS benchmark model, and an 80/20 train-test split used to validate the modelling approach. The same modelling strategy was then extended, unchanged, to a three-country comparison Spain, India, and Bangladesh chosen to span a high-income, a middle-income, and a lower-middle-income country. An Augmented Dickey-Fuller test (Dickey & Fuller, 1979) was used alongside each fitted model's own selected order to characterize the stationarity properties of each country's series. Forecast reliability was then summarized with two complementary metrics: the mean width of each country's 95% prediction interval over a 30-day horizon in absolute percentage points, and the same width expressed as a share of that country's own forecast level, to separate genuine differences in reliability from the mechanical effect of a country's coverage being closer to, or further from, saturation.

5. Results

5.1 Descriptive Overview

Figure 1 compares five countries: Spain, Germany, Czechia, Brazil, and India, chosen to span high-income European states and two larger middle- and lower-middle-income countries. Spain, Germany, and Czechia show considerably faster rollout trajectories than Brazil and India; India remains clearly below the 50% threshold throughout the observation window.

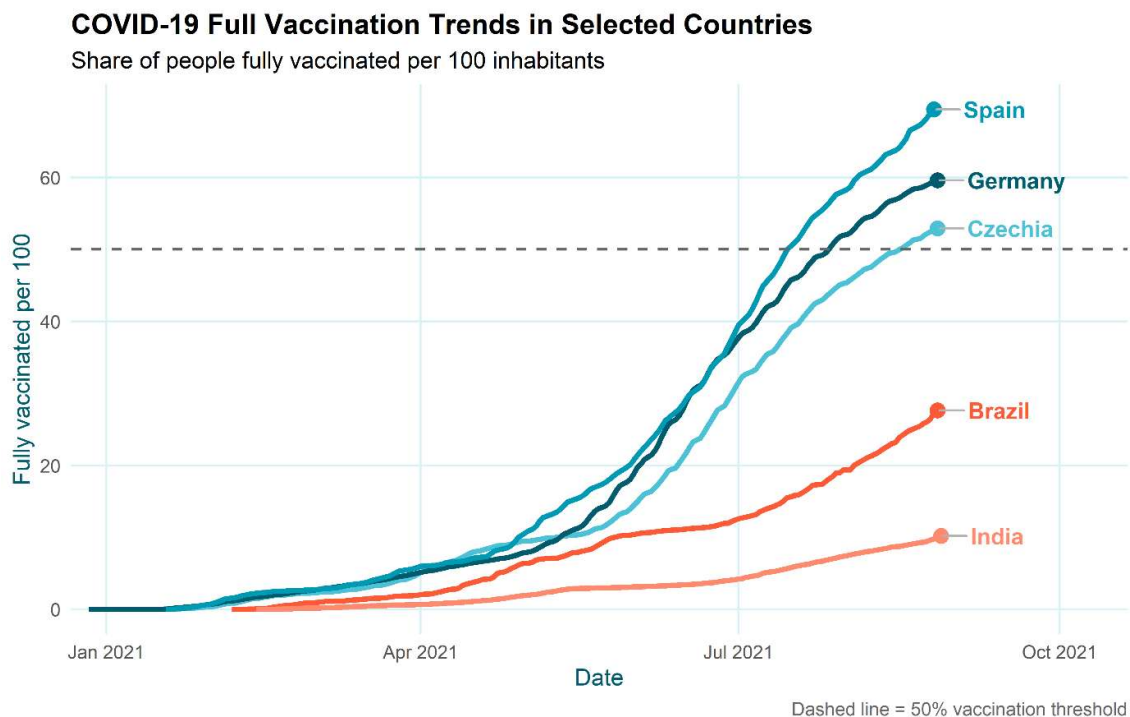


Figure 1. COVID-19 full vaccination trends in five selected countries. The dashed line marks the 50% threshold.

Figures 2 and 3 extend this comparison to the full set of 216 countries with available data. The highest-coverage country, Gibraltar, reported 116.4 fully vaccinated per 100 inhabitants at its latest observation – a figure above 100 because it reflects vaccination of non-resident cross-border workers, not an error in the data. The lowest, the Democratic Republic of the Congo, reported 0.01, a ratio of roughly 11,640-fold between the two extremes.

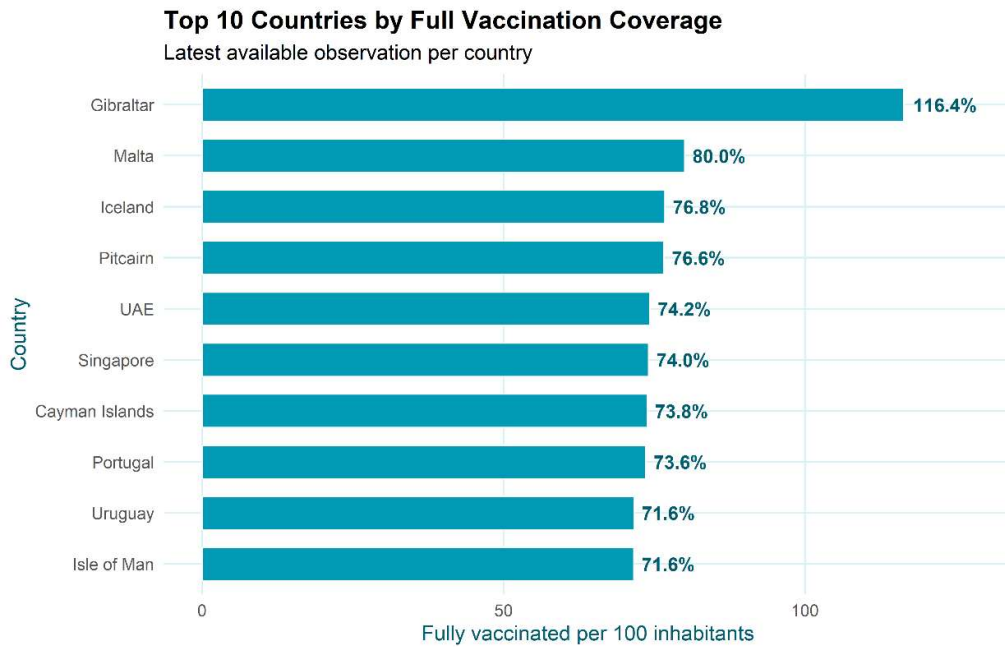


Figure 2. Top 10 countries by full vaccination coverage (latest available observation per country).

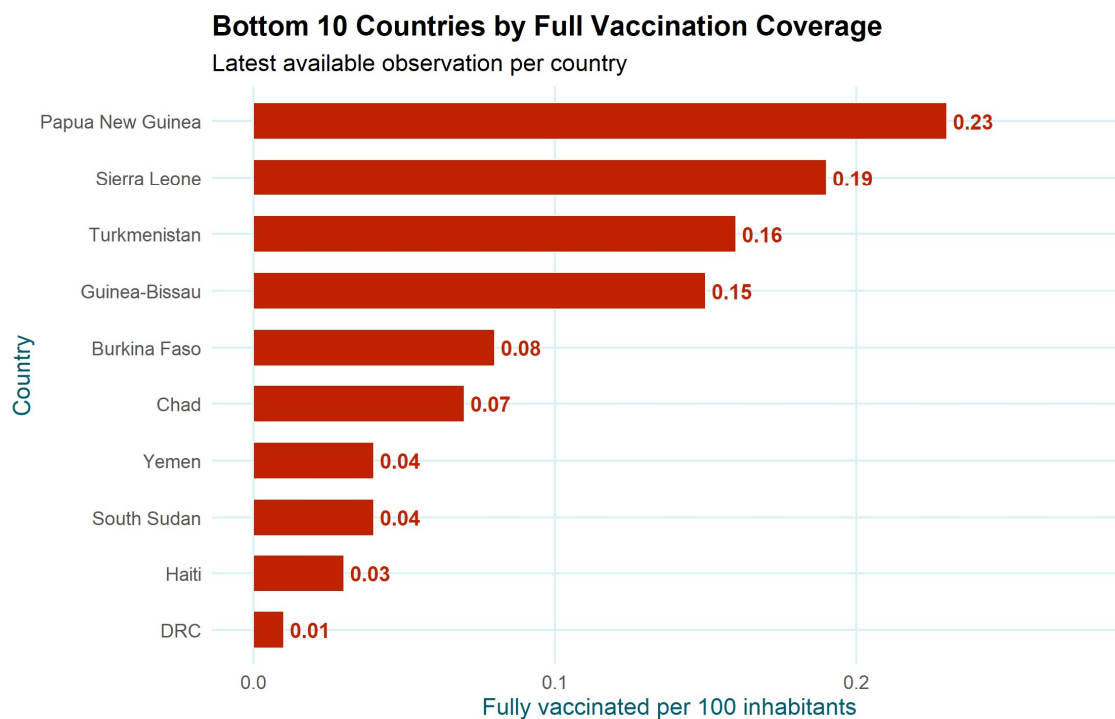
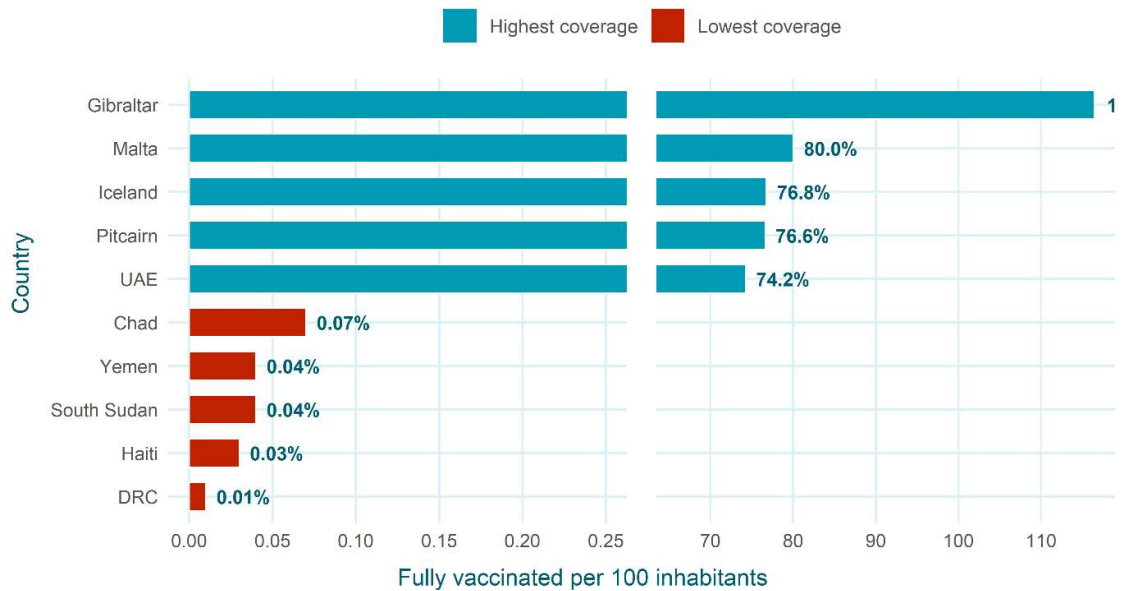


Figure 3. Bottom 10 countries by full vaccination coverage (latest available observation per country).

Because this gap spans more than three orders of magnitude, Figure 4 places the five highest- and five lowest-coverage countries on a single chart using an axis break, an axis-compression technique not covered in the course lectures, so that both extremes remain legible together rather than compressing the low end into an unreadable sliver of the chart.

Vaccination Coverage: Highest and Lowest Countries

Axis break separates the two ranges (gap between 0.25 and 66)



Source: Our World in Data COVID-19 vaccination dataset (Mathieu et al., 2021)

Figure 4. Highest- and lowest-coverage countries shown together on a broken axis.

5.2 Rollout Speed

Of the 216 countries with any reported full-vaccination data, only 61 (28.2%) reached the 50% coverage threshold at any point during the observation window; the remaining 155 (71.8%) did not. Figure 5 lists the ten fastest of those 61 countries by name: Pitcairn, Falkland Islands, Bhutan, Cook Islands, Nauru, Saint Helena, Gibraltar, Mongolia, China, and Qatar. Countries occasionally cited elsewhere for rapid rollout but not appearing in this specific top-10 list are described separately: Israel, for example, reached the threshold in 72 days well inside the 61-country median of 160 days but ranks 12th, just outside this chart.

Among the 61 countries that reached the threshold, the median time was 160 days and the mean 139 days (Figure 6); the mean falling below the median indicates a small number of unusually rapid campaigns pulling the average down, consistent with the right-skew visible in the histogram.

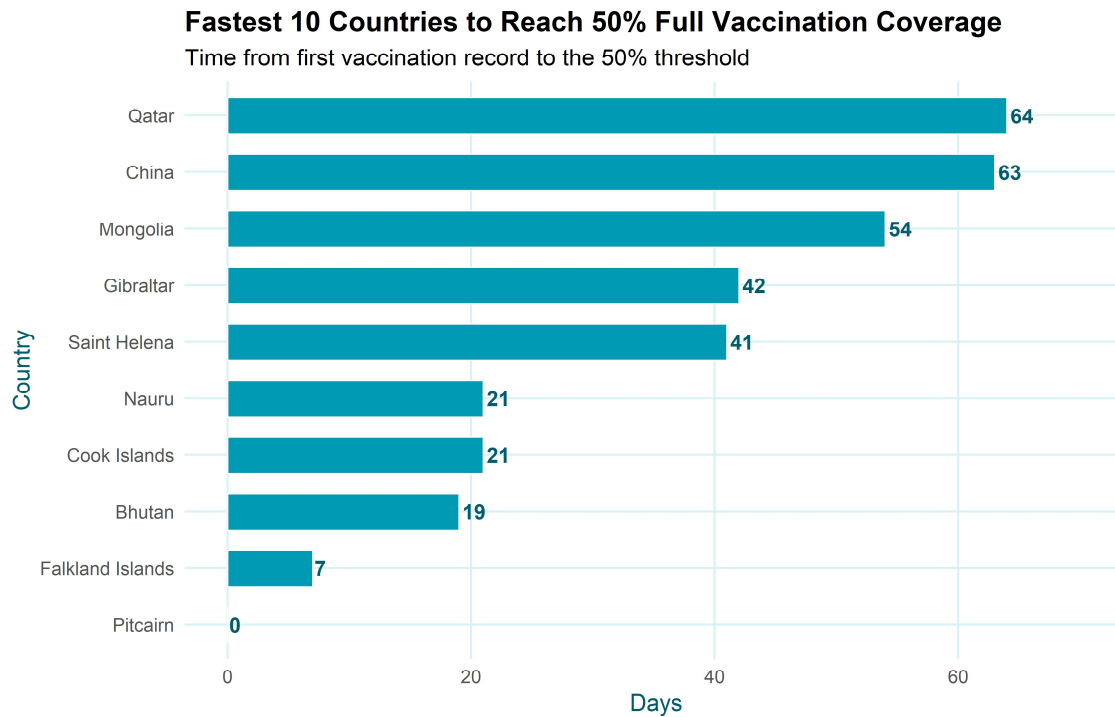


Figure 5. Fastest 10 countries to reach 50% full vaccination coverage.

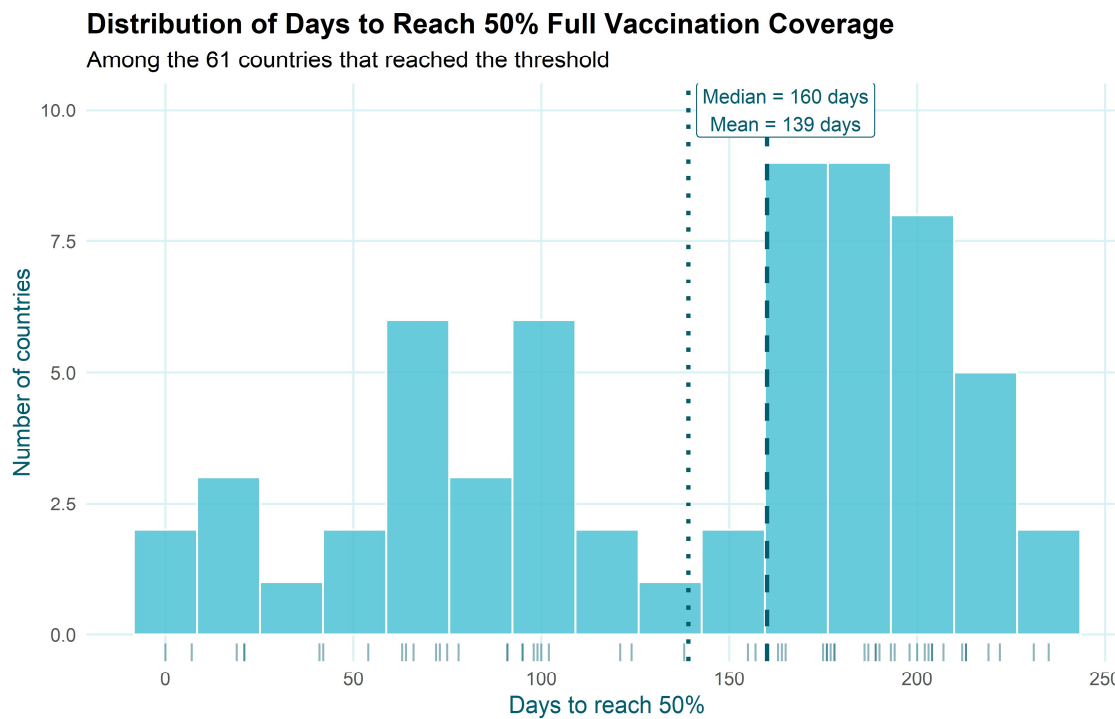


Figure 6. Distribution of days required to reach 50% full vaccination coverage, among the 61 countries that achieved it.

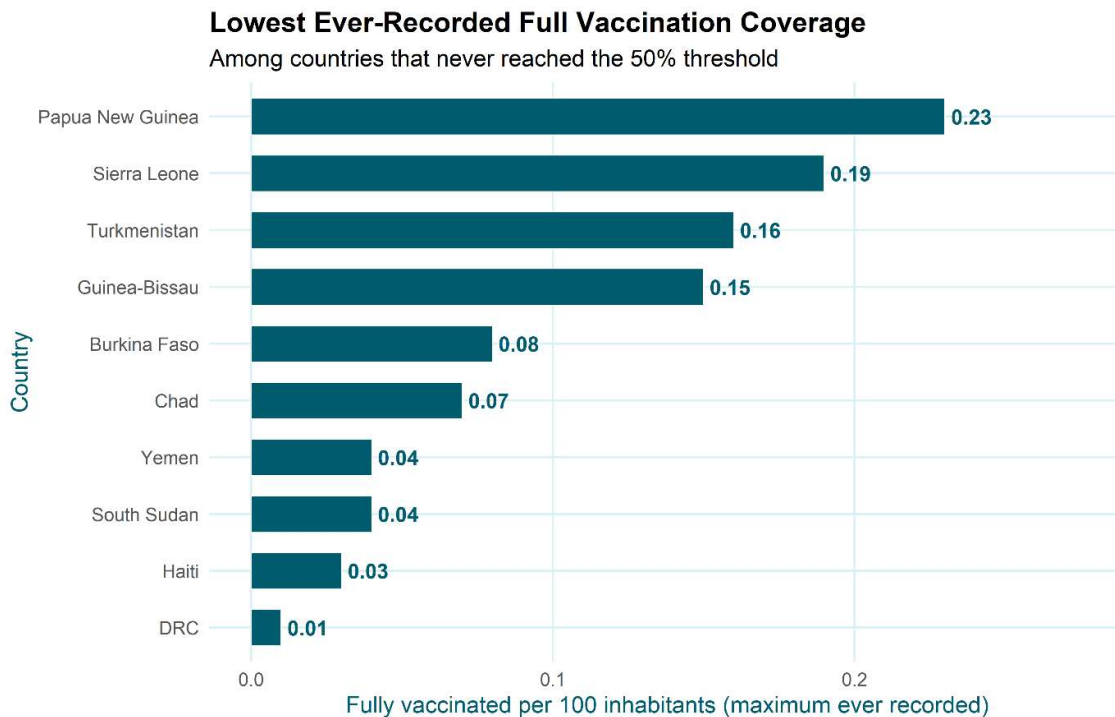


Figure 7. Lowest ever-recorded coverage among countries that never reached the 50% threshold.

Figure 7 restricts attention to the 155 countries that never reached 50%, ranking them by each country's own maximum ever-recorded coverage rather than its single latest snapshot, which is a more conservative measure for this specific group. The Democratic Republic of the Congo (0.01), Haiti (0.03), and South Sudan and Yemen (0.04 each) recorded the lowest coverage in the entire dataset.

5.3 Missing Data as a Proxy for Inequality

Thirty-six of the 216 countries have 95% or more of their full-vaccination records missing (Figure 8, restricted to the 15 highest for readability). High missingness does not necessarily indicate the absence of a vaccination campaign: Tokelau is 100% missing from this dataset yet is reported elsewhere to have achieved over 96% coverage of its eligible population (ReliefWeb, 2021). Countries with weaker administrative or digital reporting infrastructure are systematically underrepresented here, meaning the inequalities quantified in this report are, if anything, an underestimate missing data is itself a signal of structural disadvantage, not a neutral gap (Mbousou et al., 2019).

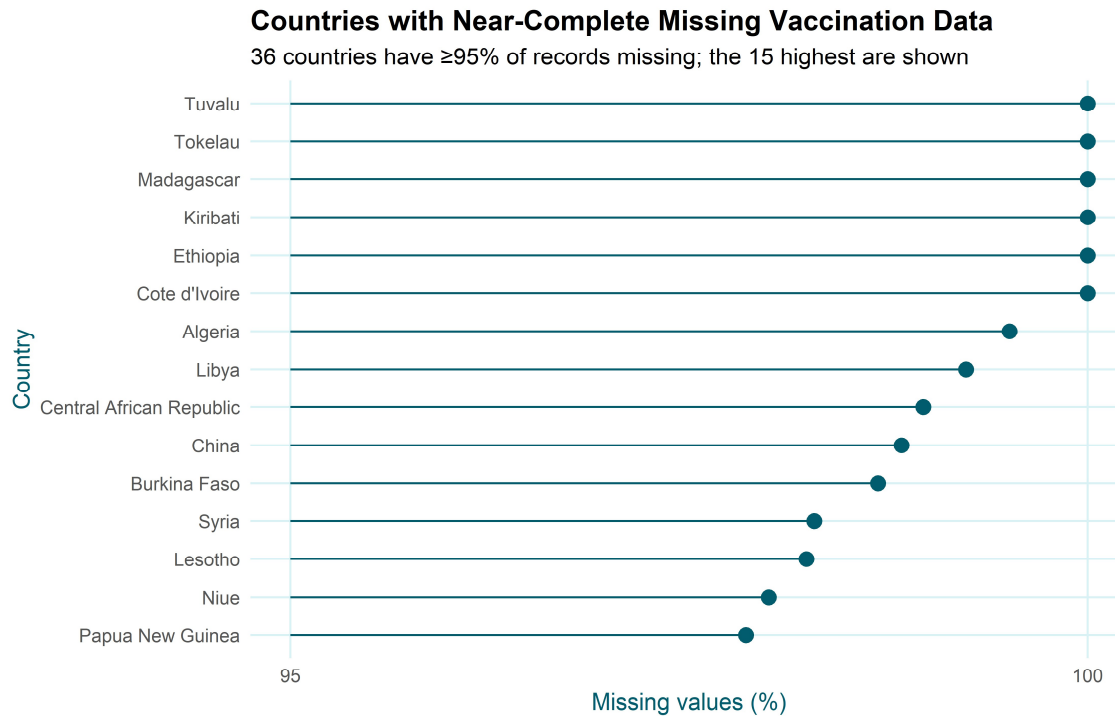


Figure 8. Countries with near-complete missing vaccination data ($\geq 95\%$ of records missing; 15 highest shown).

5.4 Geographic Patterns

Figures 9 and 10 present the same underlying inequality from two complementary geographic angles: overall coverage, and the speed at which that coverage was reached. High coverage in Figure 9 is concentrated in Europe, parts of the Americas, and isolated high-income states in Asia and Oceania; the African continent shows predominantly low shading or grey (no data). Figure 10 shows the same pattern for rollout speed, though grey on this map conflates two different things: the majority of countries that never reached the threshold at all, and countries with no usable data, which is itself a limitation of a single-colour "no data" category.

Full Vaccination Coverage by Country

Latest available observation per country — grey = no data reported

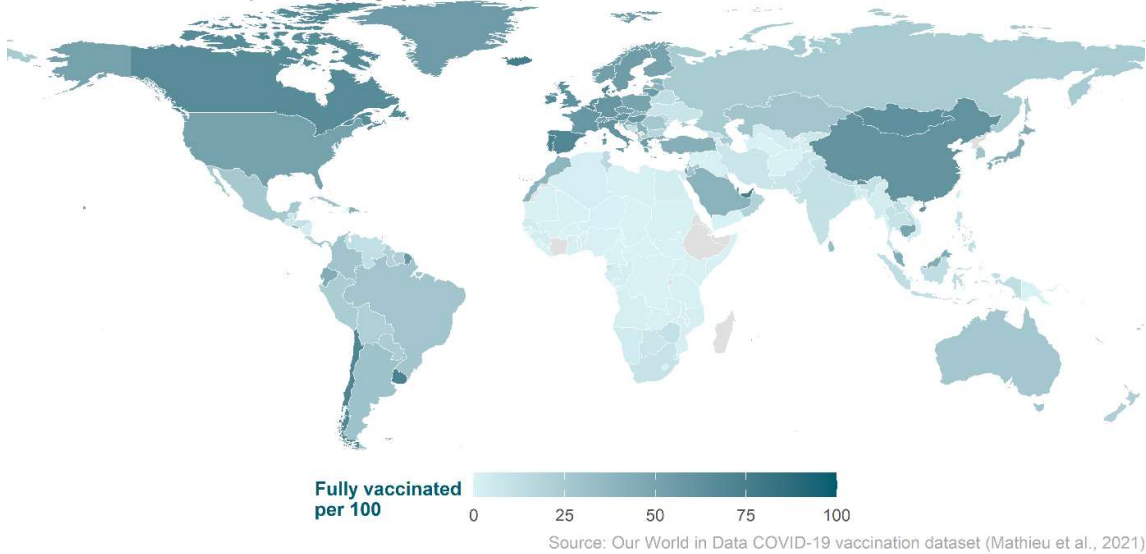


Figure 9. World map of full vaccination coverage by country (latest available observation).

Speed of Vaccination Rollout by Country

Days from first recorded vaccination to 50% coverage — grey = threshold never reached, or no data

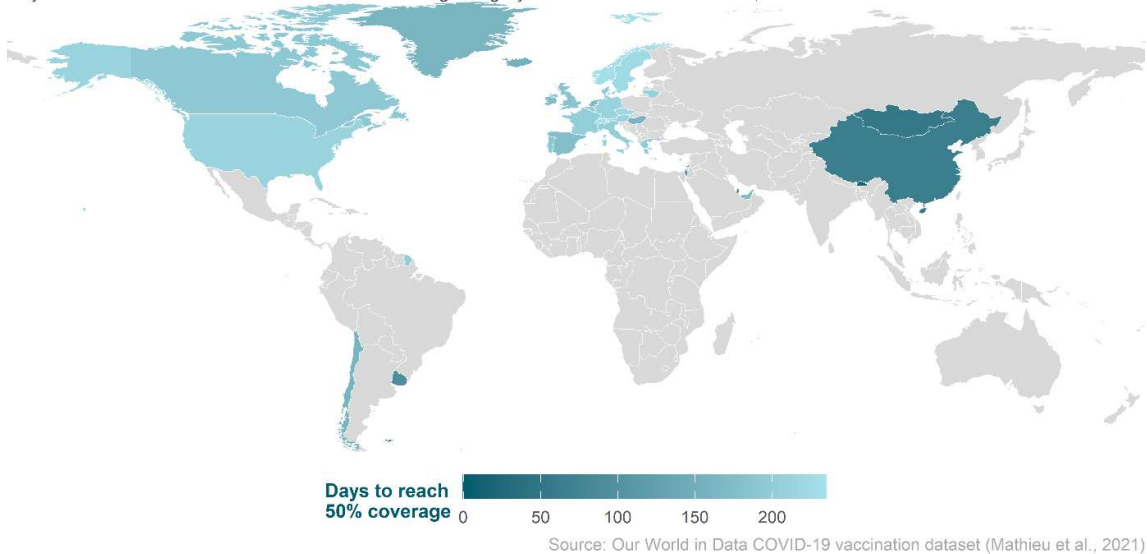


Figure 10. World map of vaccination rollout speed by country (days to reach 50% coverage).

5.5 Income-Group and Continental Inequality

The income-group time series (Figure 11) shows a gap between income groups that widens continuously across the observation period, accelerating from roughly April 2021 onward. By August 2021, high-income countries had reached approximately 52 fully vaccinated per 100 inhabitants, versus roughly 0.5 for low-income countries a difference of more than 50 percentage points using World Bank income classifications already embedded in the dataset.

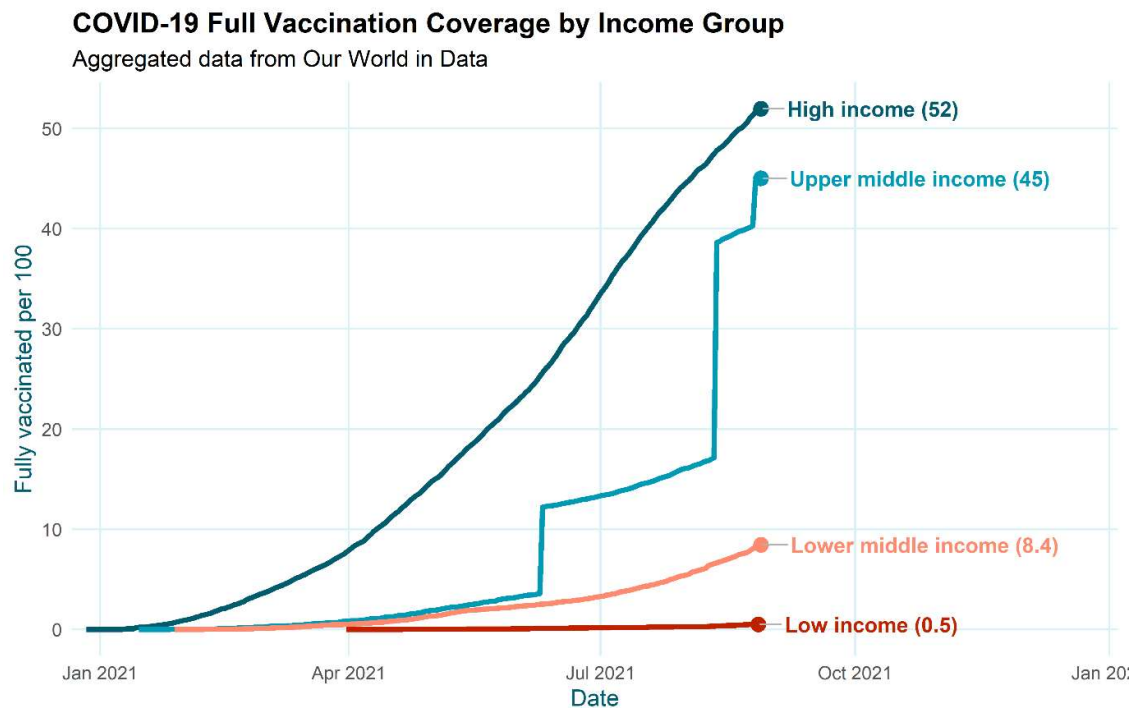


Figure 11. COVID-19 full vaccination coverage by World Bank income group over time.

The continent-level comparison (Figure 12) confirms this pattern at finer geographic resolution, combining a smoothed density, a boxplot, and individual country points in a single raincloud plot (`ggdist` and `ggbeeswarm`). One outlier, Gibraltar, was excluded from this specific figure because its coverage above 100% would compress the rest of the distribution; that exclusion is stated in the figure's own subtitle rather than left silent. Africa shows the lowest median coverage and a right-skewed distribution; Europe the highest median and the narrowest spread; Asia and the Americas both show wide within-continent variation, indicating that continental groupings alone understate the heterogeneity driven by income and infrastructure differences.

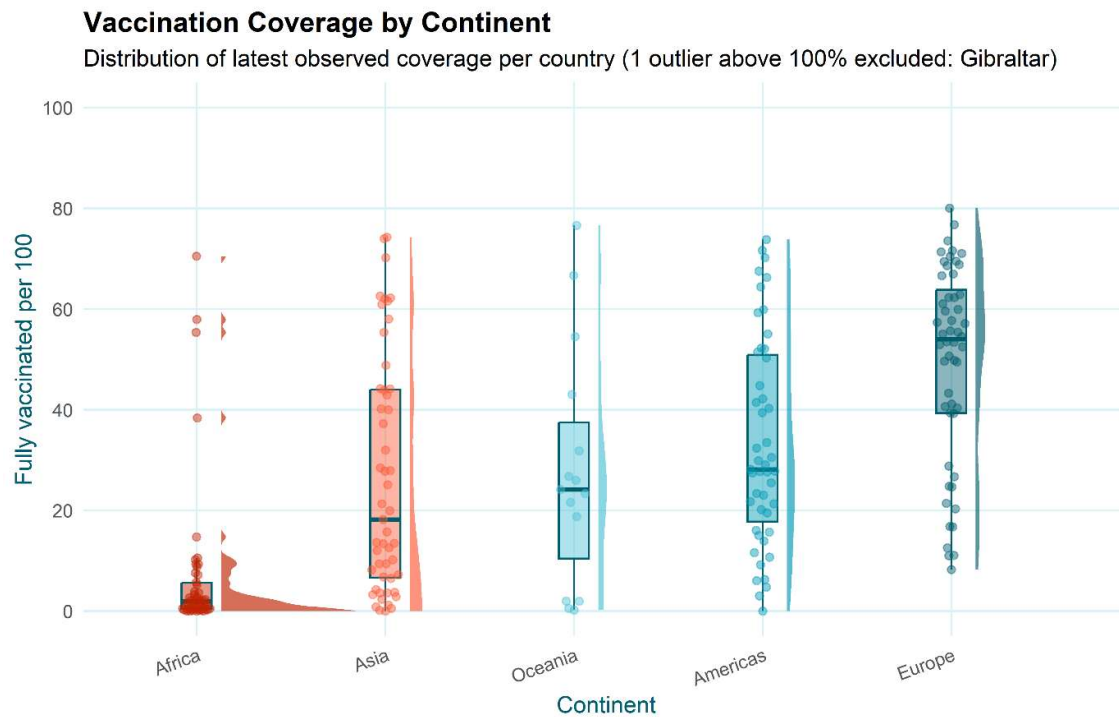


Figure 12. Distribution of vaccination coverage by continent (latest observation per country; one outlier above 100% excluded).

5.5 Correlation Analysis

Strong positive correlations ($r = 0.94\text{--}0.98$) were observed among the three cumulative vaccination indicators, confirming that countries performing well on one cumulative measure tend to perform well on all of them. Daily vaccinations per million showed only weak-to-moderate correlation with the cumulative measures ($r = 0.16\text{--}0.40$), indicating that a country's vaccination activity at its most recent reporting date was largely decoupled from its overall campaign success. Countries that had already completed their rollout were no longer administering high daily volumes, while countries early in rollout could show high daily rates relative to their still-low cumulative progress.

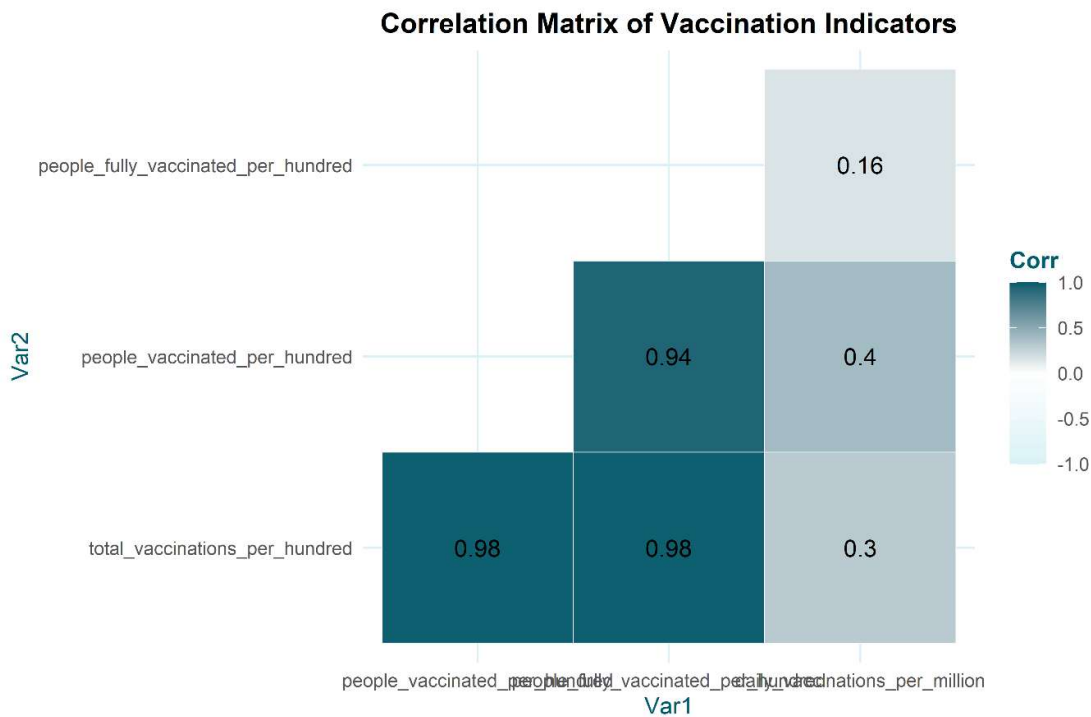


Figure 13. Correlation matrix of vaccination indicators (latest observation per country).

5.6 PCA and K-means Clustering

Principal component analysis was computed on 209 of the 216 countries with complete data across all four indicators; seven were excluded for missing at least one indicator (Bonaire Sint Eustatius and Saba, Bulgaria, Eswatini, Hungary, Moldova, the Philippines, and Turkmenistan). The first component explained 76.5% of total variance and the second 22.8%, together accounting for 99.3% matching, rather than contradicting, the figures reported for the same computation elsewhere in this report. PC1 separates countries primarily by overall vaccination coverage, with lower-income African and Asian countries clustering at one end and European countries at the other (Figure 14); the loadings plot (Figure 15) confirms that the three cumulative indicators load together on PC1, while daily vaccinations per million loads mostly on the near-orthogonal PC2.

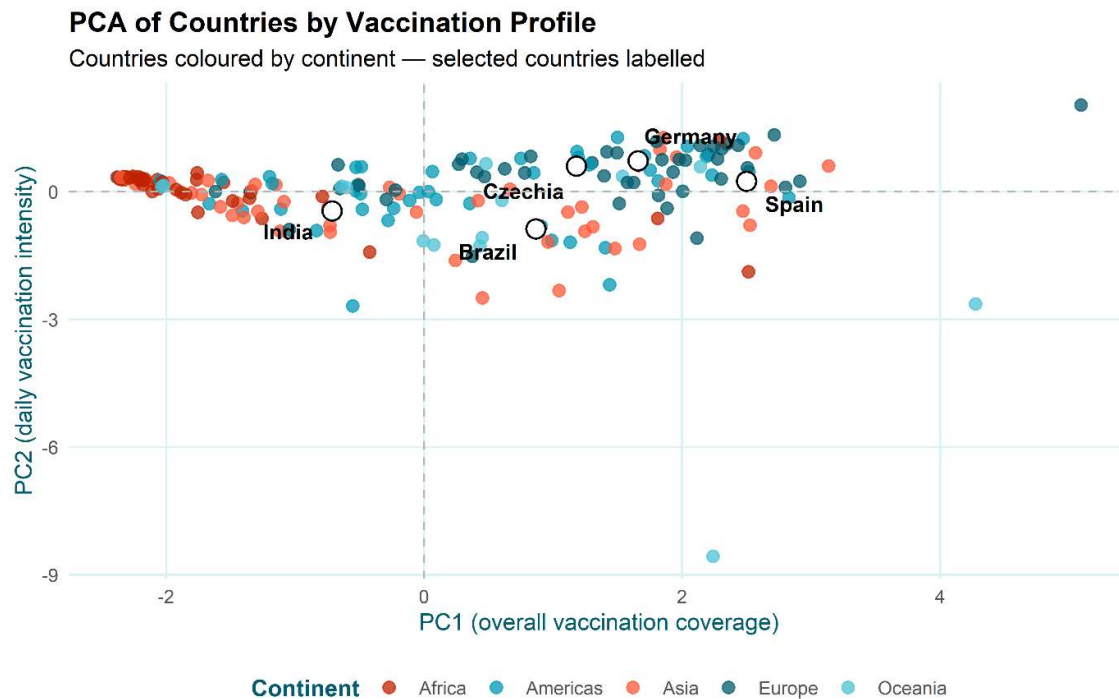


Figure 14. PCA of countries by vaccination profile, coloured by continent.

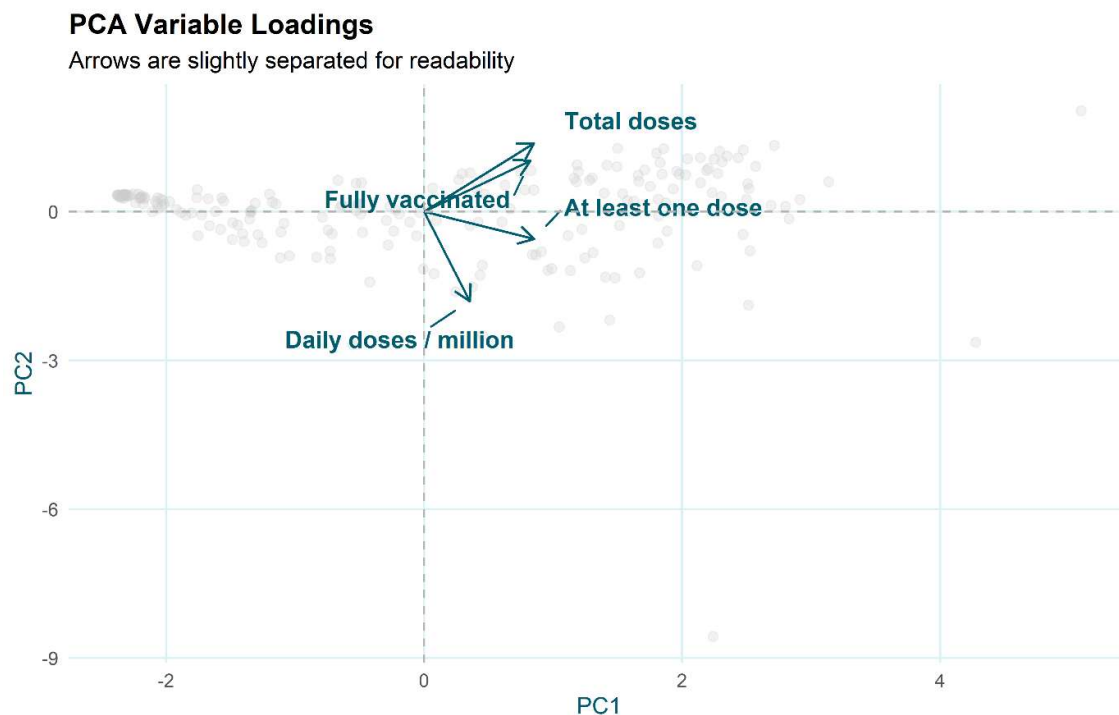


Figure 15. PCA variable loadings.

Before fixing the number of clusters, both the elbow method and the average silhouette width were computed (Figure 16); together with the theoretically expected high/moderate/low segmentation, they support retaining $k = 3$.

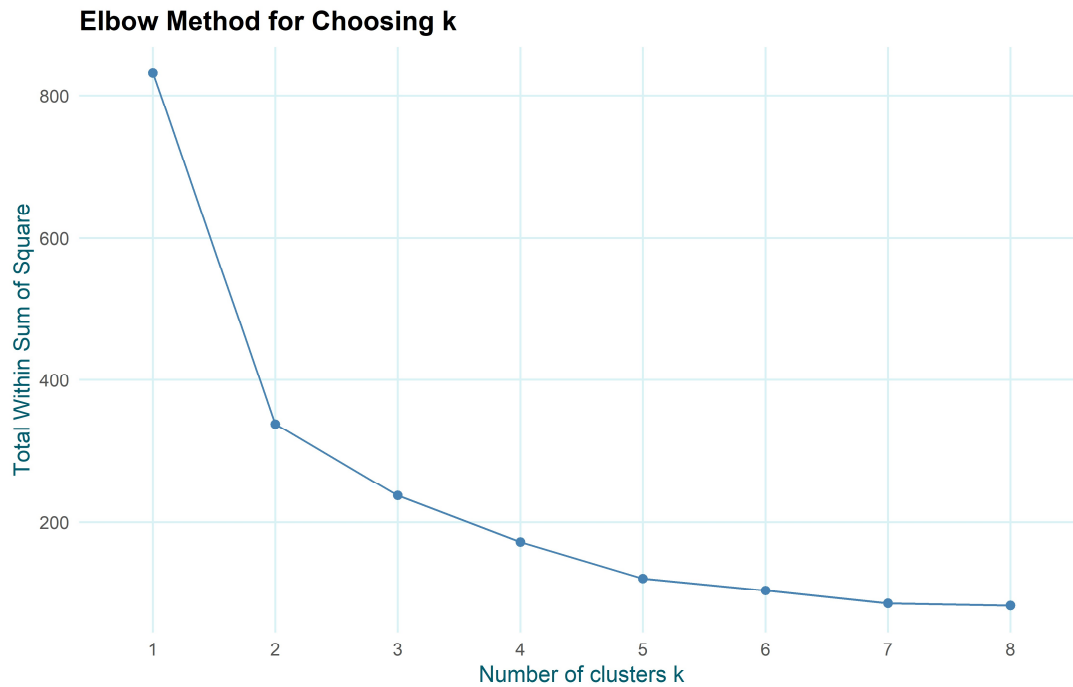


Figure 16a. Elbow method (within-cluster sum of squares) for choosing k .

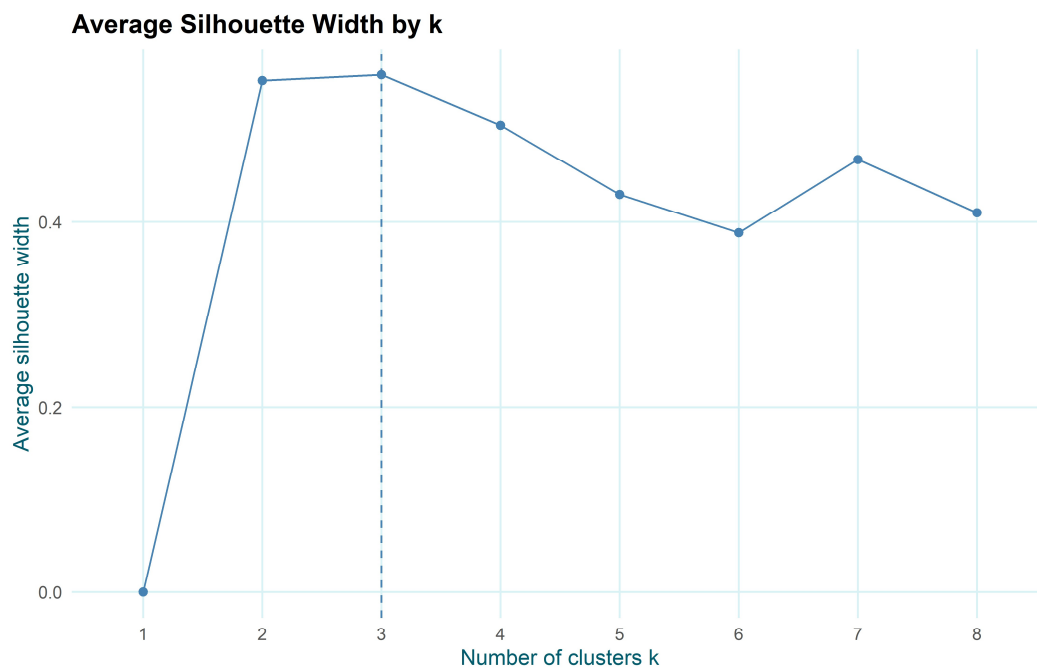


Figure 16b. Average silhouette width by k .

K-means clustering with $k = 3$ (Figure 17) identifies three structurally distinct groups, labelled from each cluster's own computed mean coverage rather than an arbitrary index: a high-coverage profile of 70 countries (mean 59.7 fully vaccinated per 100), a moderate-coverage profile of 35

countries (mean 32.7), and a low-coverage profile of 104 countries (mean 8.9). Spain, Germany, and Czechia fall in the high-coverage cluster; Brazil in the moderate-coverage cluster; India in the low-coverage cluster consistent with the trajectories already shown in Figure 1. The clear geometric separation between clusters, rather than a continuous gradient, suggests that vaccination inequality in this period was structural and categorical rather than smoothly graded

K-means Clusters in PCA Space (k = 3)

Cluster labels assigned from each cluster's own computed mean coverage

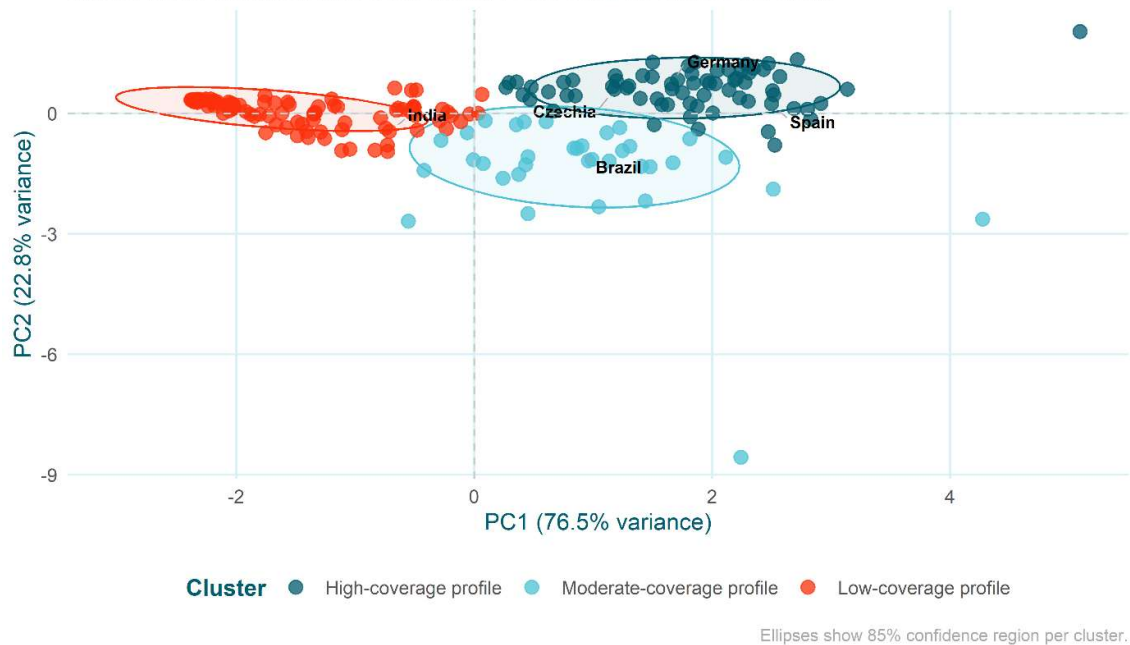


Figure 17. K-means clusters in PCA space (k = 3); labels assigned from each cluster's own computed mean coverage.

5.7 ARIMA Forecasting

Germany's coverage series, re-expressed on a complete daily calendar (no gaps were present) and modelled with `auto.arima()`, selected an ARIMA(2,2,1)(2,1,0)[7] specification. Residual diagnostics were adequate: the Ljung-Box test returned $Q^* = 13.72$ on 9 degrees of freedom ($p = 0.133$), indicating no significant remaining autocorrelation.

On an 80/20 train-test split, the ARIMA model outperformed an ETS benchmark on the held-out test set (RMSE 2.44 vs. 3.05; MAE 1.69 vs. 2.15), supporting ARIMA as the more suitable model for this series.

ARIMA Forecast: Germany Full Vaccination Coverage

30-day forecast with 80% and 95% confidence intervals

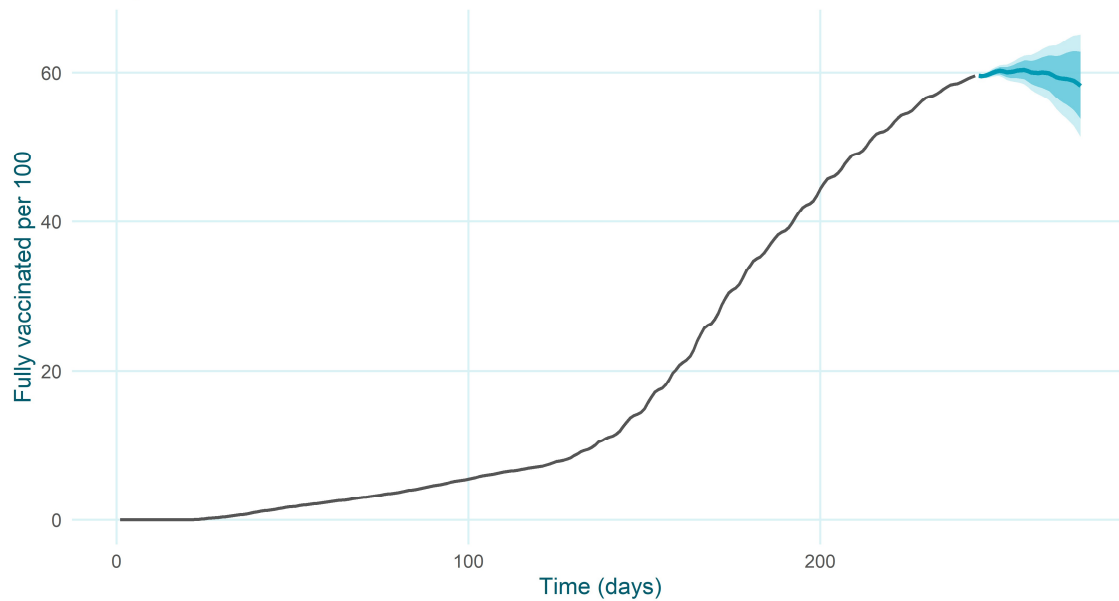


Figure 18. ARIMA forecast of Germany's full vaccination coverage (30-day horizon, 80% and 95% prediction intervals).

Residuals from ARIMA(2,2,1)(2,1,0)[7]

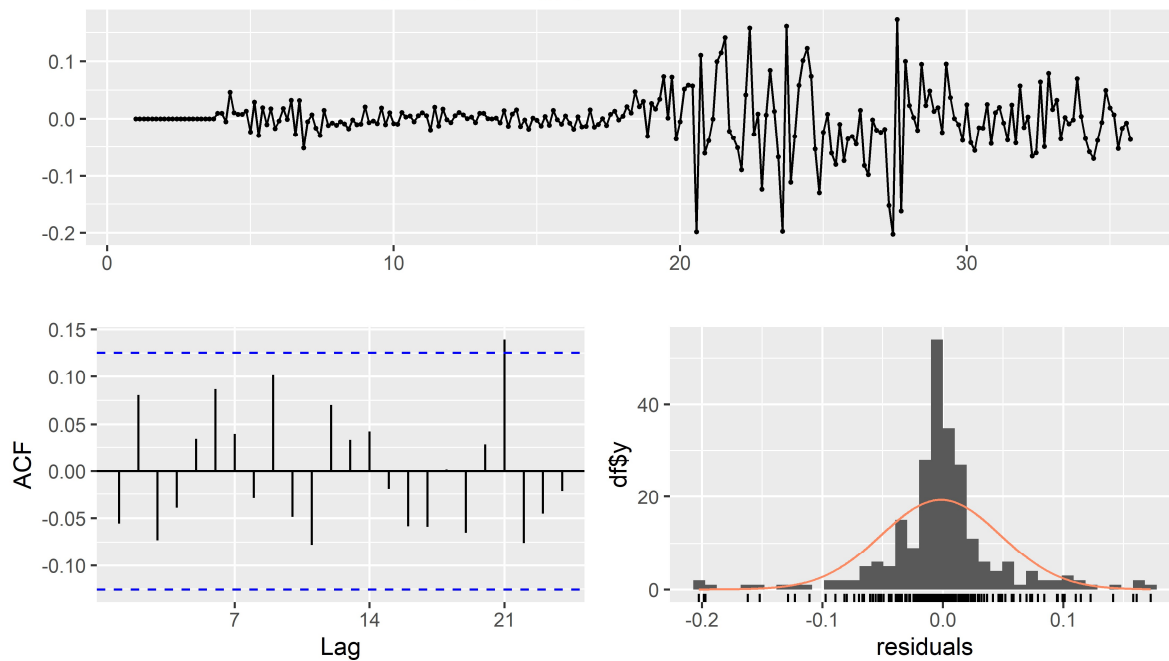


Figure 19. Residual diagnostics for Germany's ARIMA model.

ETS Forecast: Germany Full Vaccination Coverage

30-day forecast with 80% and 95% confidence intervals

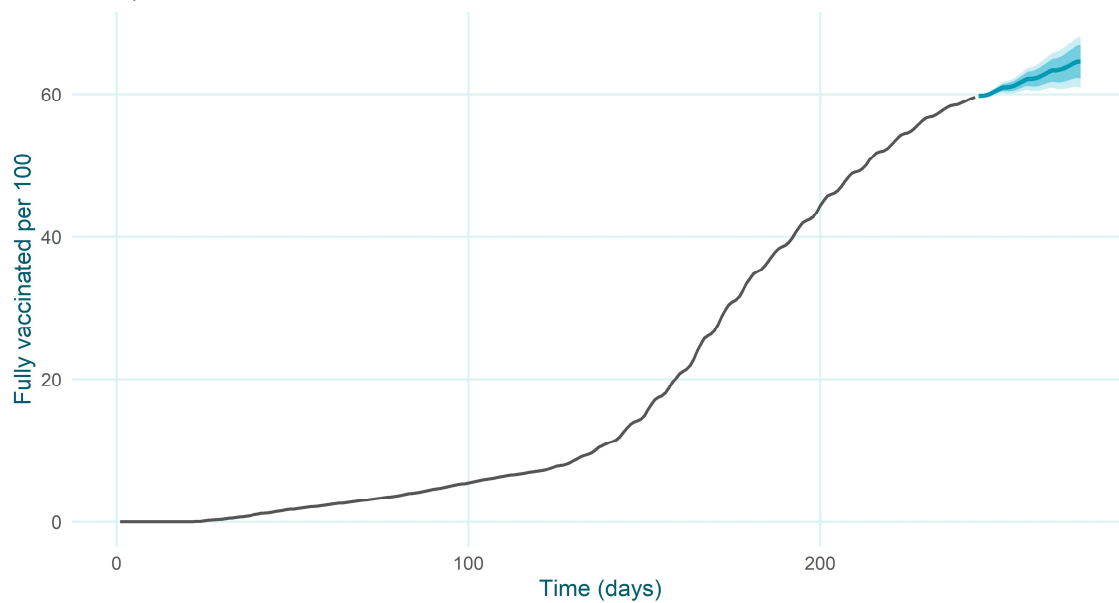


Figure 20. ETS forecast of Germany's full vaccination coverage, shown for comparison.

The same modelling strategy auto-selected differencing applied to a gap-filled level series, with no manual pre-differencing was then extended to Spain, India, and Bangladesh (Figure 21). Spain's model selected order $(1,1,1)(1,1,1)[7]$ with an Augmented Dickey-Fuller p-value of 0.559 on levels (consistent with a unit root, supporting differencing); India selected $(0,2,2)(2,0,0)[7]$ ($p = 0.990$); Bangladesh selected $(0,2,3)(2,0,0)[7]$ ($p = 0.951$).

ARIMA Forecast of Full Vaccination Coverage: Spain

30-day forecast with 80% and 95% confidence intervals

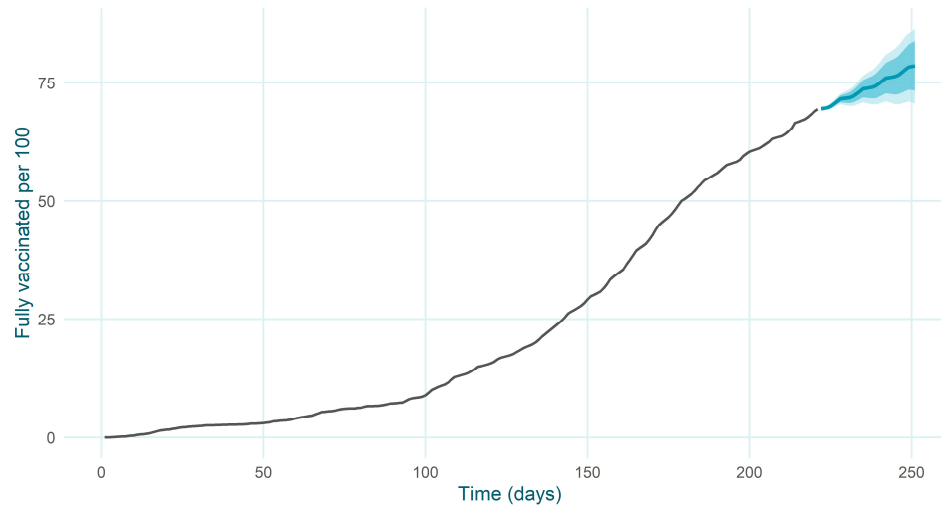


Figure 21a. ARIMA forecast of full vaccination coverage: Spain.

ARIMA Forecast of Full Vaccination Coverage: India

30-day forecast with 80% and 95% confidence intervals

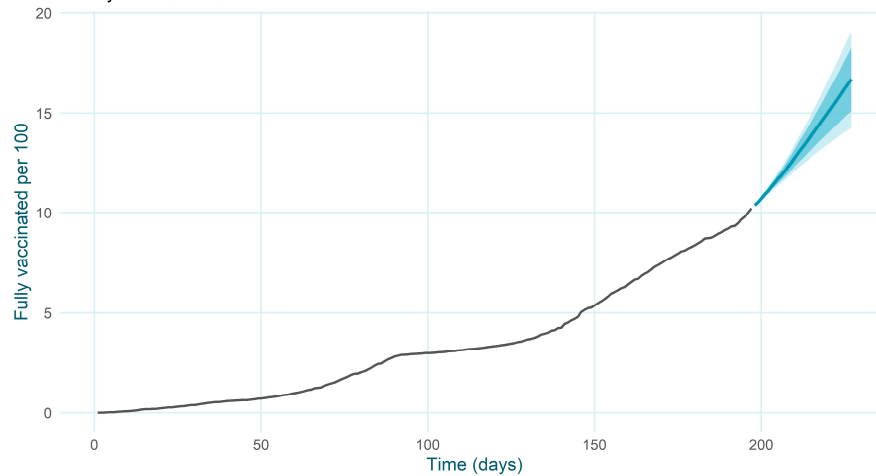


Figure 21b. ARIMA forecast of full vaccination coverage: India.

ARIMA Forecast of Full Vaccination Coverage: Bangladesh

30-day forecast with 80% and 95% confidence intervals

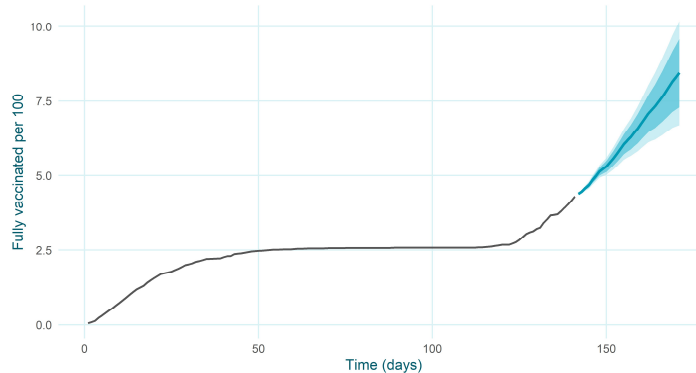


Figure 21c. ARIMA forecast of full vaccination coverage: Bangladesh.

The two forecast-uncertainty metrics in Figure 22 do not tell the same story, and reporting both rather than only the more convenient one is the central methodological point of this section. In absolute terms (Figure 22a), Spain's mean 95% prediction-interval width over the 30-day horizon was the widest of the three countries (7.04 percentage points), compared with 1.99 for India and 1.41 for Bangladesh the opposite of the naive expectation that a high-income country with more complete data should produce the narrowest, most reliable forecast. This is best explained by scale: Spain's model required full seasonal differencing ($D = 1$), which compounds uncertainty faster over a multi-week horizon, and Spain's coverage was already close to saturation (around 70%) by the end of the series, where absolute week-to-week swings simply have more room to occur than in a series still rising from near zero.

Once each country's interval width is rescaled by its own forecast level (Figure 22b), the ranking reverses and lines up with the original hypothesis: Bangladesh's forecast is proportionally the most uncertain relative to its own trajectory (22.4% of its forecast level), followed by India (14.9%), with Spain proportionally the most precise (9.5%). Read together, absolute and relative uncertainty answer two different questions how wide is the band in real terms, and how uncertain is the forecast relative to what is actually being forecast and only the second of these supports treating forecast reliability as unequally distributed by income level once the scale confound is removed.

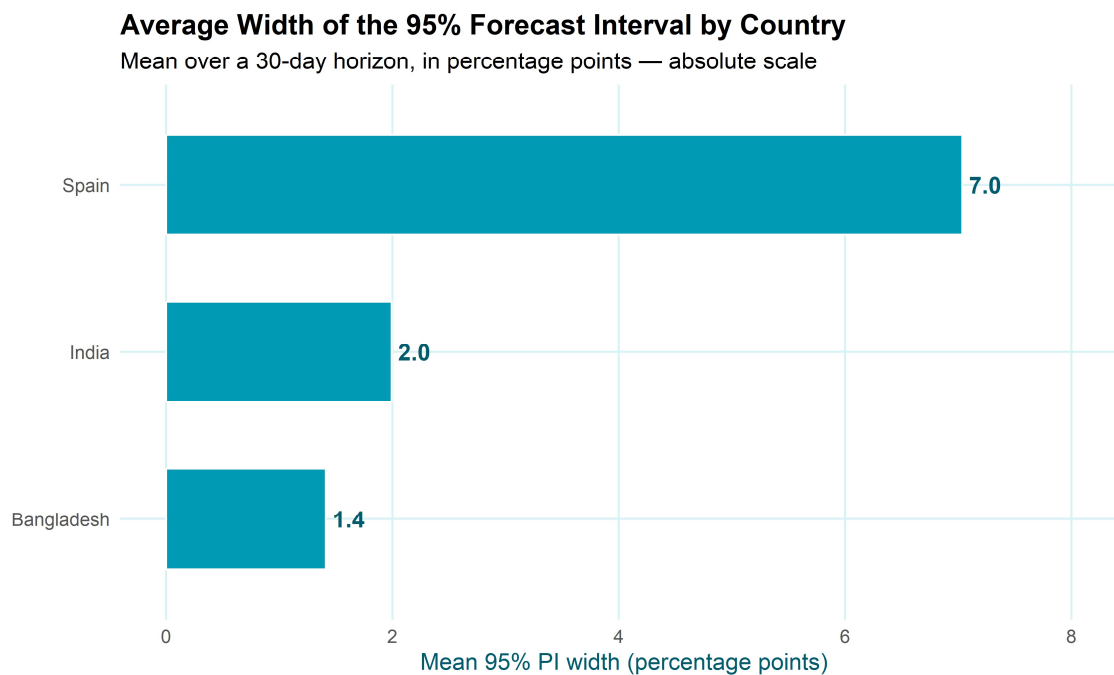


Figure 22a. Mean 95% prediction-interval width by country, in absolute percentage points.

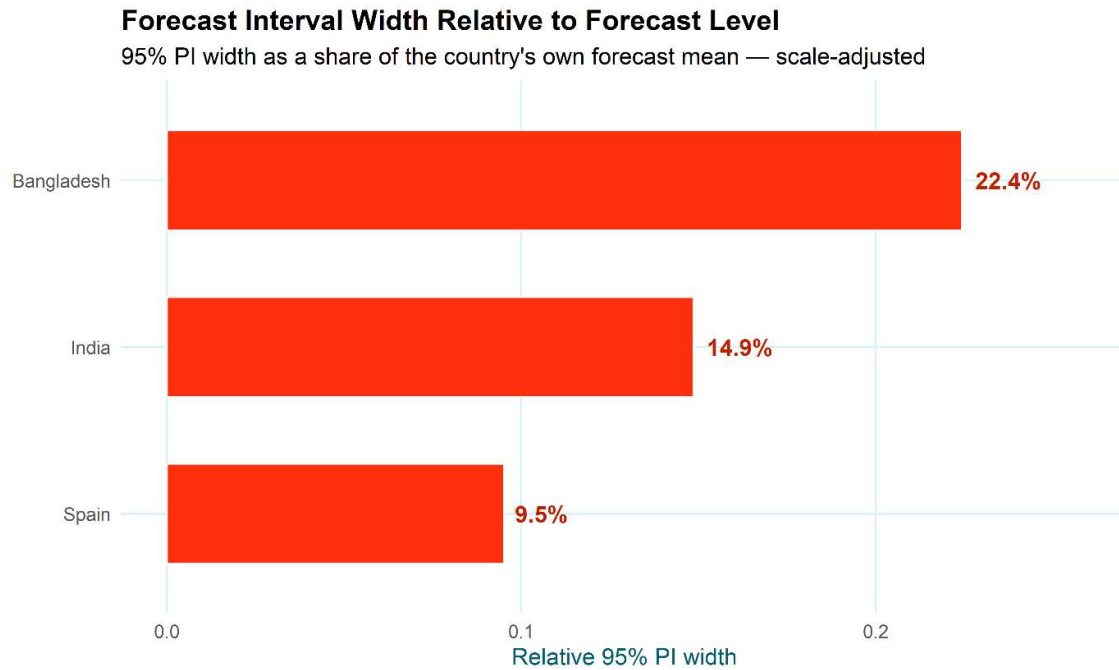


Figure 22b. Mean 95% prediction-interval width by country, relative to each country's own forecast level.

6. Discussion

6.1 Strengths

This analysis combines descriptive statistics, unsupervised learning, and time series forecasting to characterize COVID-19 vaccination inequality from cross-sectional, longitudinal, and predictive perspectives using a single, publicly available, regularly updated dataset. The World Bank income-group aggregates already embedded in the dataset were used directly, avoiding the need for an external cross-walk between country names and income classifications. The ARIMA component adds a forward-looking dimension that goes beyond description: by testing, rather than assuming, whether forecast uncertainty differs by income level and by reporting both the absolute result that contradicted the initial hypothesis and the scale-adjusted result that supported it the analysis demonstrates that inequality can be a structural feature of the data-generating process itself, not only of the outcomes it measures.

6.2 Limitations

Several limitations must be acknowledged. First, and most importantly, cross-sectional comparisons based on each country's "latest observation" (Figures 2, 3, 12, 14, and 17) are not all observed on the same calendar date: these dates range from April to August 2021, and at least 15 of the 216 countries have a snapshot more than 30 days older than the true end of the observation window, meaning their true position by late August 2021 could be higher than shown here. Second, as demonstrated in Section 5.3, near-complete missingness does not necessarily mean the absence of vaccination activity, so the coverage gap quantified in this report may itself be an underestimate for the most data-poor countries. Third, the observation period ends in August 2021, capturing only the first phase of the global rollout; the substantial expansion of vaccination programmes in lower-income countries during 2022 and 2023 falls outside this analysis. Fourth, the principal component analysis excluded seven countries with incomplete indicator data, a small (3%) but non-random subset. Finally, the ARIMA forecast-uncertainty comparison in Section 5.8 illustrates a broader methodological caution: an absolute metric and a scale-adjusted metric can point in opposite directions, and neither should be reported alone when the countries being compared differ as much in their baseline coverage level as Spain, India, and Bangladesh do.

6.3 Implications

These findings have direct implications for public health policy and global health governance. The concentration of early vaccine access in high-income countries, combined with systematic under-reporting from lower-income countries, underscores the need for both more equitable vaccine distribution frameworks and stronger investment in the health-data infrastructure needed to monitor them. The forecast-reliability finding adds a further, more specific implication: agencies or organizations that rely on short-term forecasting models to plan vaccine allocation or logistics should expect and explicitly account for systematically wider relative uncertainty in exactly the countries where planning support is most needed, rather than treating forecast unavailability or imprecision as a neutral gap.

7. Conclusions

COVID-19 vaccination trends differed substantially across countries between December 2020 and August 2021. High-income countries reached higher coverage earlier and more consistently than lower-income countries; only 28.2% of countries reached the 50% full-vaccination threshold at all, and among those that did, time-to-threshold ranged from 0 to 235 days. Missing-data patterns, the PCA and clustering structure, and the ARIMA forecast-uncertainty comparison all point in the same direction: inequality in this period was structural, and it extended beyond vaccination outcomes into the reporting infrastructure and predictive tools used to monitor them. The one qualification worth restating plainly is that this last claim survives only once forecast uncertainty is measured on a scale that accounts for each country's own coverage level the naive, unscaled comparison actually pointed the other way, and reporting that inversion honestly is itself part of this report's conclusion, not a complication to be hidden from it.

Future work could extend this analysis to the full 2021–2023 period to capture later phases of the global rollout, once comparably clean daily data become available. Integrating additional covariates gross domestic product per capita, healthcare expenditure, and COVAX participation status would allow a more granular analysis of the determinants of vaccine access, and applying the same absolute-versus-relative forecast-uncertainty framework used here to a larger set of countries would test whether the scale-adjusted pattern found for Spain, India, and Bangladesh generalizes more broadly.

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